
MATH 493

Algebra

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1 Groups

1.1 Groups and Group Representations

Definition 1.1.1. A group G is a set with a binary operation s.t.

$$(a, b) \rightarrow a \cdot b$$

- Associativity: $(a \cdot b) \cdot c = a(b \cdot c)$
- Identity: $\exists e \in G$ s.t. $a \cdot e = a = e \cdot a$
- Inverse: Given $a \in G$, $\exists b \in G$ s.t. $ab = e = ba$

Example 1.1.2. • Show that the identity is unique

- If $ab = ac$, then $b = c$

- *Proof.* Let e_1, e_2 be identities in G

$$e_1 = e_1 e_2 = e_1$$

□

- *Proof.* $ab = ac$

$$\exists a^{-1} \text{ s.t. } a^{-1}a = e$$

$$a^{-1}(ab) = a^{-1}(ac) = b = c$$

□

Example 1.1.3. • $\{e\}$

- $(\mathbb{Z}, +)$
- $(\mathbb{Q}^\times, \cdot)$
- $\mathbb{Z} / n\mathbb{Z}$ (Cyclic Groups)

$$n \in \mathbb{Z} \text{ if } a, b \in \mathbb{Z} \text{ are congruent with } n \text{ if } n \mid (a - b)$$

Definition 1.1.4. Write $[a]$ as the equivalence class of a .

$$[a] + [b] = [a + b]$$

$$\text{Check if } [a] = [a'] \text{ and } [b] = [b'] \implies [a + b] = [a' + b']$$

Abelian: commutative

- S_3 Permutations of $\{1, 2, 3\}$

$$- (1) = e$$

- (12) = σ
- (13) = $\tau\sigma = \sigma\tau^2$
- (23) = $\sigma\tau$
- (123) = τ
- (132) = τ^2

top first	e	σ	$\sigma\tau$	$\sigma\tau^2$	τ	τ^2
e	e	σ	$\sigma\tau$	$\sigma\tau^2$	τ	τ^2
σ	σ	e	τ	τ^2	$\sigma\tau$	$\sigma\tau^2$
$\sigma\tau$	$\sigma\tau$	τ^2	e	τ	$\sigma\tau^2$	σ
$\sigma\tau^2$	$\sigma\tau^2$	τ	τ^2	e	σ	$\sigma\tau$
τ	τ	$\sigma\tau^2$	σ	$\sigma\tau$	τ^2	e
τ^2	τ^2	$\sigma\tau$	$\sigma\tau^2$	σ	e	τ

This is the same as D_3

$$e = \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix}$$

$$\sigma = \begin{bmatrix} -1 & 0 \\ 0 & 1 \end{bmatrix}$$

$$\tau = \begin{bmatrix} \cos(\theta) & -\sin(\theta) \\ \sin(\theta) & \cos(\theta) \end{bmatrix} \theta = 120$$

$$\tau^2 = \begin{bmatrix} \cos(\theta) & -\sin(\theta) \\ \sin(\theta) & \cos(\theta) \end{bmatrix} \theta = 240$$

$$\sigma\tau = \begin{bmatrix} -\cos(\theta) & \sin(\theta) \\ \sin(\theta) & \cos(\theta) \end{bmatrix} \theta = 120$$

$$\sigma\tau^2 = \begin{bmatrix} -\cos(\theta) & \sin(\theta) \\ \sin(\theta) & \cos(\theta) \end{bmatrix} \theta = 240$$

1.2 Subgroups

Definition 1.2.1. Let G be a group then $H \leq G$ if

- Contains same Identity
- Closed under group operation
- Closed under taking inverses

Definition 1.2.2. The order of a group, denoted $|G|$ is the cardinality of G

Definition 1.2.3. The order of $a \in G$, the order of a is the smallest $n \in \mathbb{N}$ s.t. $a^n = e$

Theorem 1.2.4 (Lagrange's Theorem). *Let G be a finite group. Let $H \leq G$ then $|H|$ divides $|G|$*

Corollary 1.2.5. *Let $a \in G$ then $\text{ord}(a)$ divides $|G|$*

Definition 1.2.6. Let $a \in G$ and $H \leq G$ then the left coset $aH = \{ah \mid h \in H\}$

Example 1.2.7. $\mathbb{Z}_n$ and let $H = \langle 2 \rangle$

$$a = 0 \text{ "}aH\text{"} = a + H = \langle 2 \rangle$$

$$a = 1 \{1, 3\}$$

$$a = 2 \{0, 2\}$$

$$a = 3 \{1, 3\}$$

Proposition 1.2.8.

1. Any two left cosets are either identical or disjoint
2. G is the disjoint union of the left cosets of H in G

1.3 Normal Subgroups, Quotient groups and the first isomorphism theorem

Definition 1.3.1. A subgroup $H \subset G$ is said to be normal in G if left coset of H in G is also a right coset.

Proposition 1.3.2.

1. H is normal in G
2. $aH = Ha \forall a \in G$
3. $aHa^{-1} = H \forall a \in G$
4. $aHa^{-1} \subseteq H \forall a \in G$
5. $H \subset aHa^{-1} \forall a \in G$

Theorem 1.3.3. Suppose $\varphi : G_1 \rightarrow G_2$ is a group homomorphism then there is a unique isomorphism $\psi : G_1 / \ker \varphi \rightarrow \text{im} \varphi$ s.t. the diagram commutes

$$\begin{array}{ccc}
 G_1 & \xrightarrow{\varphi} & \text{im}(\varphi) \leq G_2 \\
 \downarrow \pi & \nearrow \psi & \\
 G_1 / \ker \varphi & &
 \end{array}$$

Commutates: $\psi(g \ker \varphi) = \varphi(g)$